Building Firewall

420-521-VA ADVANCED NETWORKS

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GitHub Repository: https://github.com/daiyan03samin/firewall.git

Date: 10/11/2026

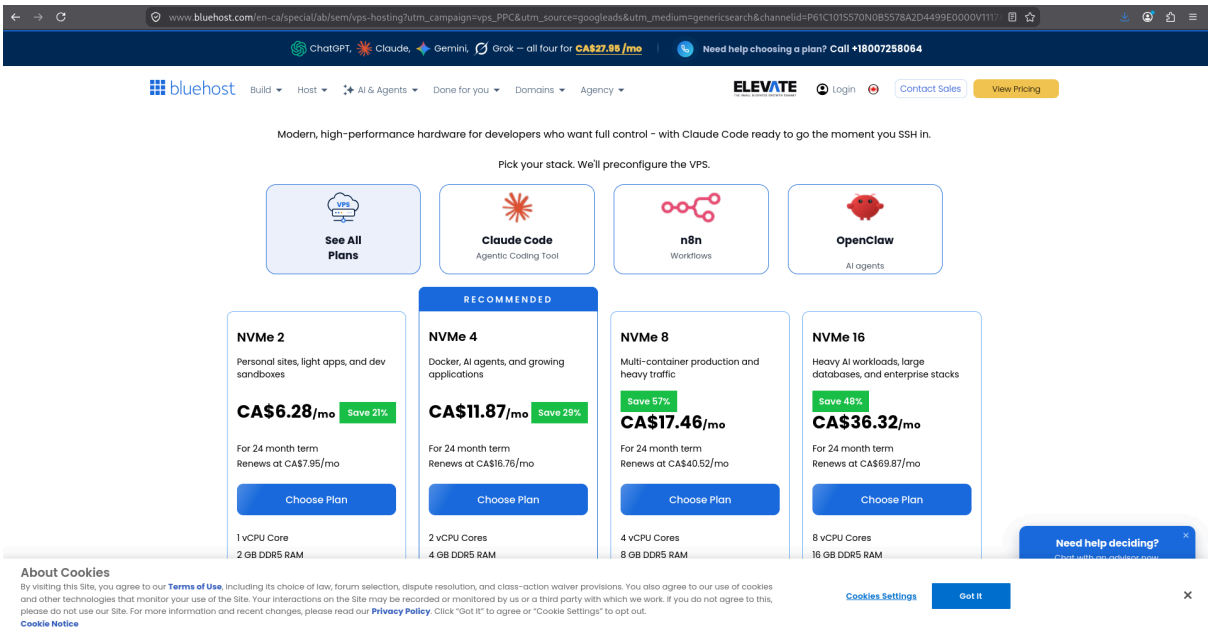
Aaron Hossain Daiyan Abrar Samin

Aaron’s Work

VPS Setup

Research

Researching possible VPS providers for the project:

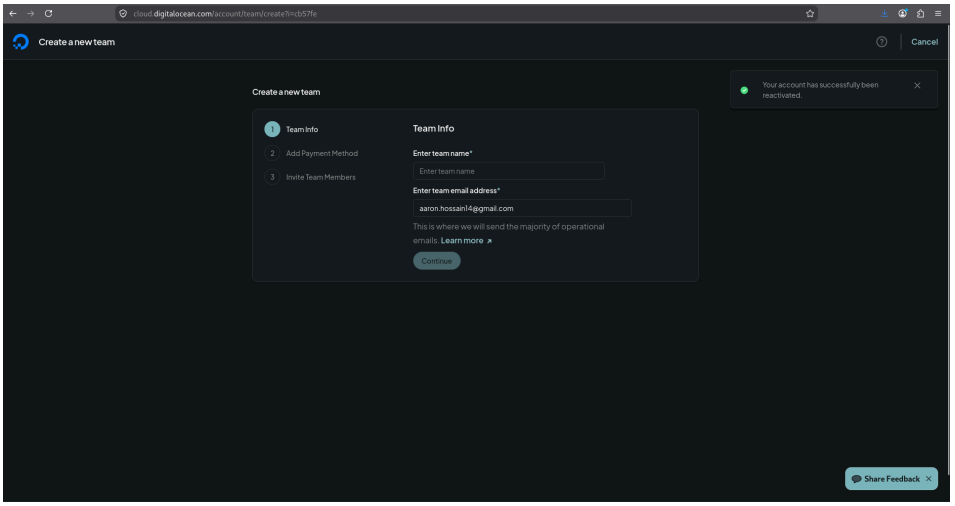


I’m looking at this one that I have never heard of: the prices are respectable, so it seems like a suitable option.

However, last semester (for eCommerce), I used DigitalOcean for the VPS.

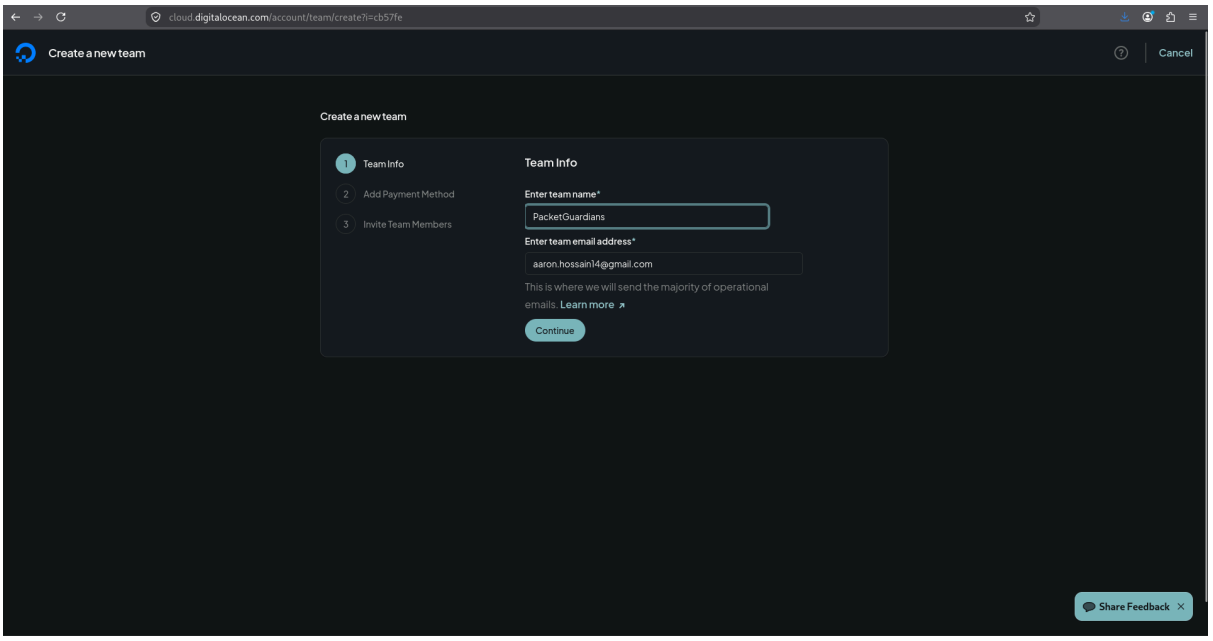
This one is for 24 months though, which is much longer than I need it for. So i will probably use DigitalOcean again for simplicity's sake.

Reactivating my account on DigitalOcean



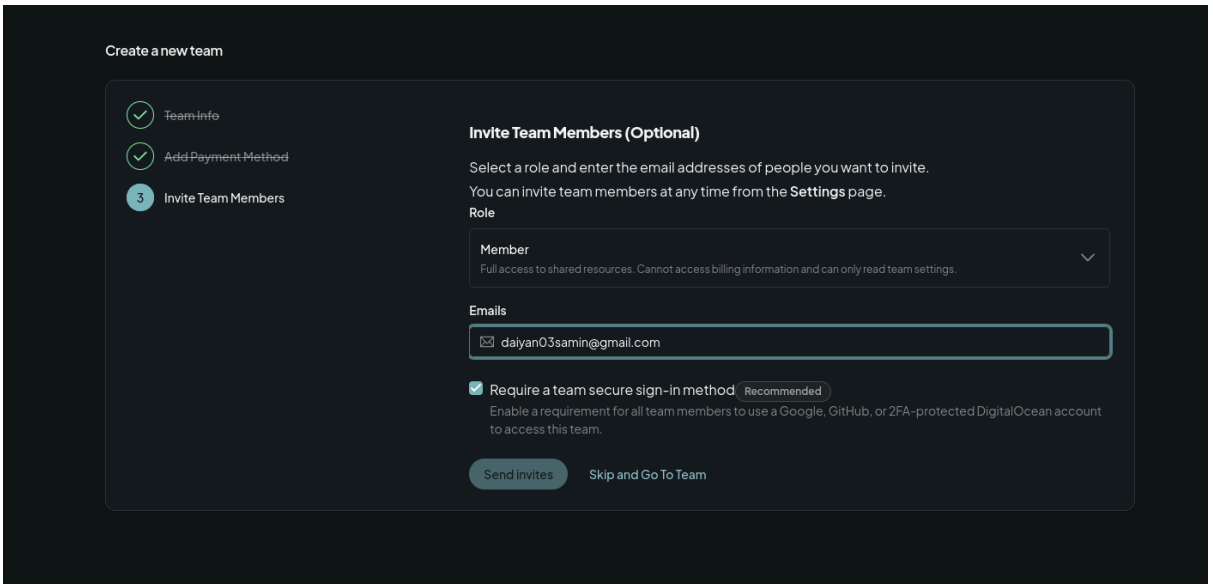
Since my account was inactive for over a year, DigitalOcean deactivated it and I had to reactivate it.

Setting up the VPS



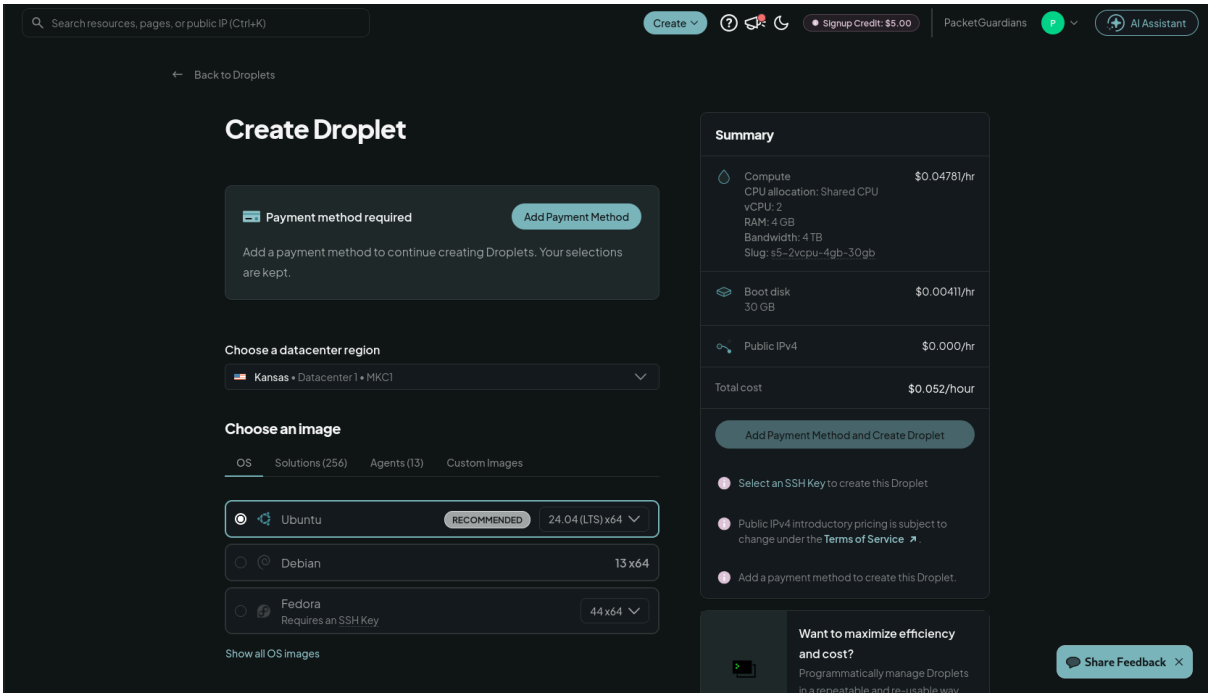
I started creating a DigitalOcean team named PacketGuardians and entered my email address to receive operational notifications.

Inviting team member to the team



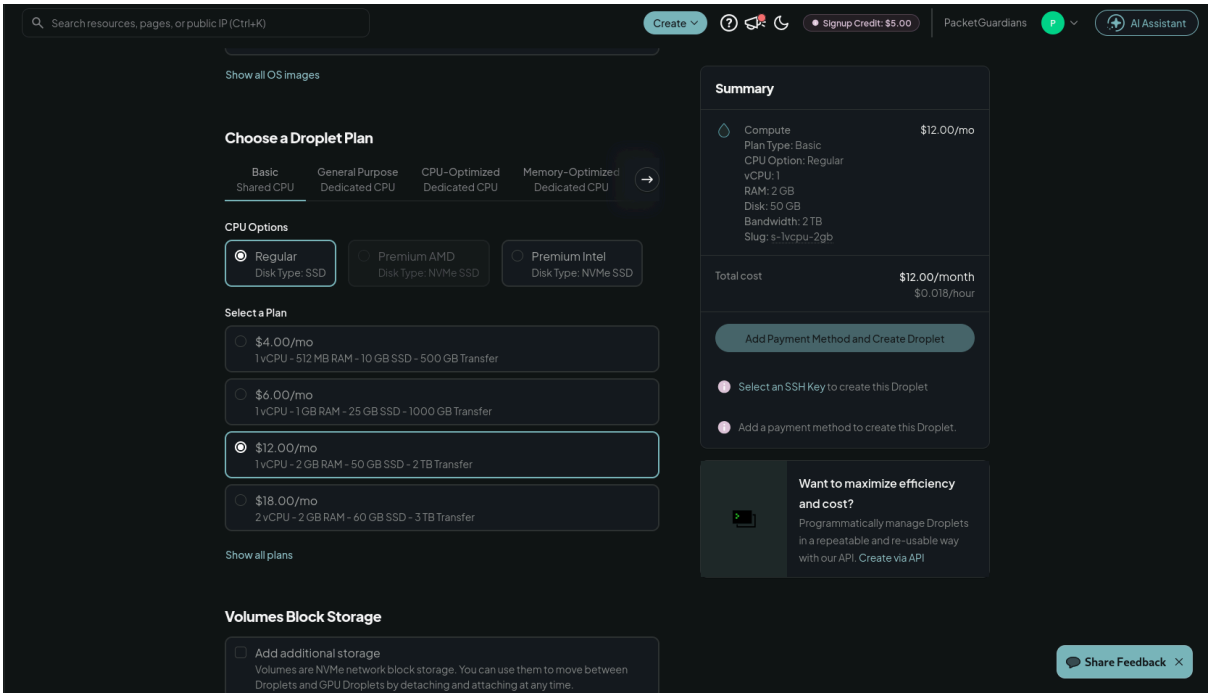
Here, I'm adding my teammate, Daiyan, to the DigitalOcean team and granting him full access by giving him the ‘Member’ role. With this, he will be able to set up the firewall rules and policies with me.

Creating a droplet (VPS)



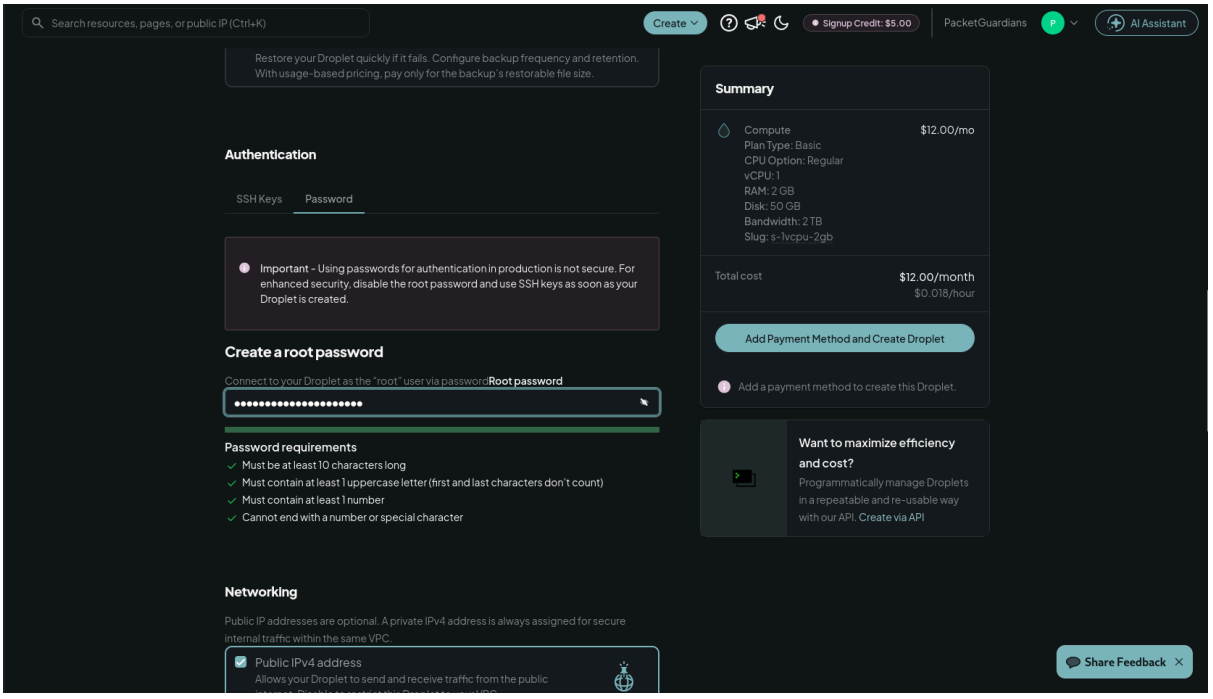
Here, I'm setting up the actual VPS (DigitalOcean calls it droplets). I looked through the datacenter regions and a lot of them had the basic plans unavailable so I had to switch it to San Francisco Datacenter 2 (SFO2).

Choosing the droplet (VPS) plan



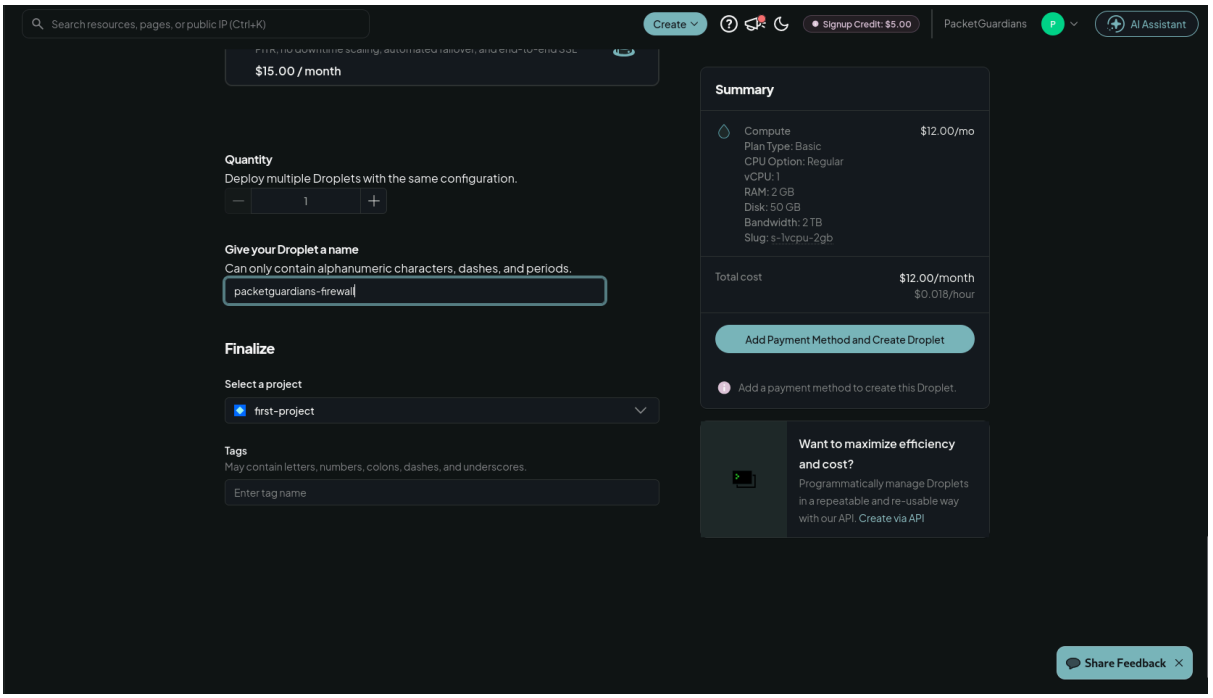
I selected the Basic plan that comes with 1vCPU, 2GB RAM, 50GB SSD and 2TB transfer. After doing some research (using AI too), this setup should be sufficient for our firewall project.

Creating a password for the droplet



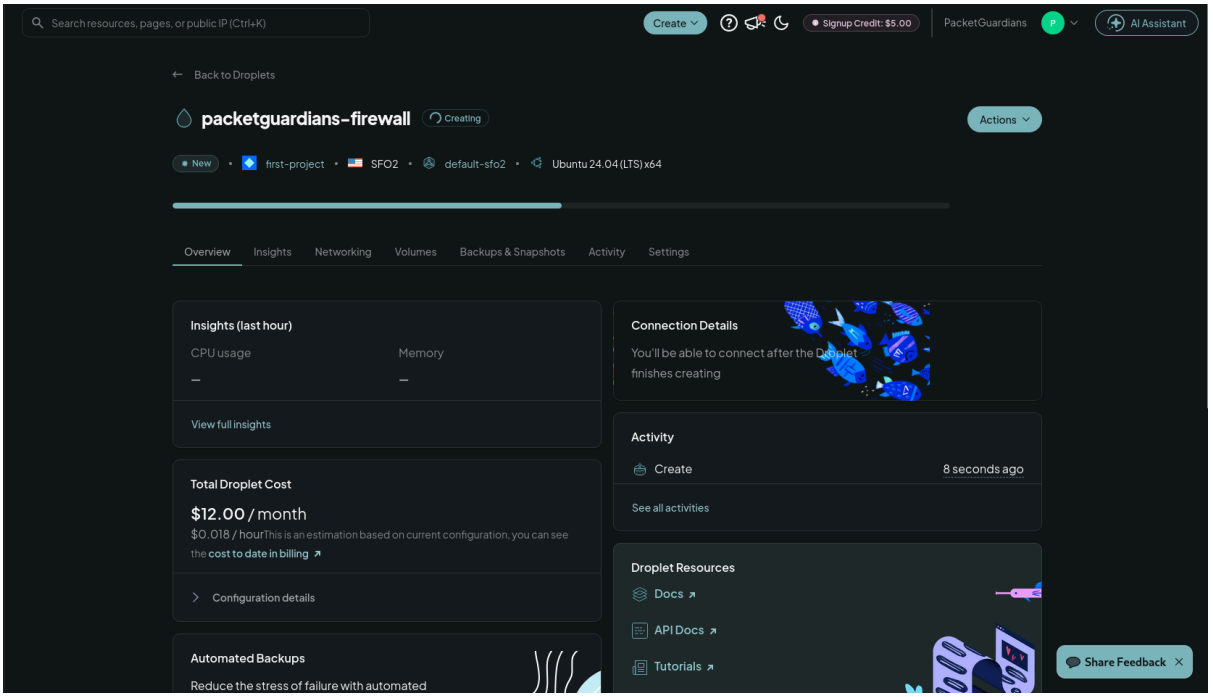
Here, I decided to use a password for the VPS instead of the SSH key to keep things fairly simple.

Naming the droplet



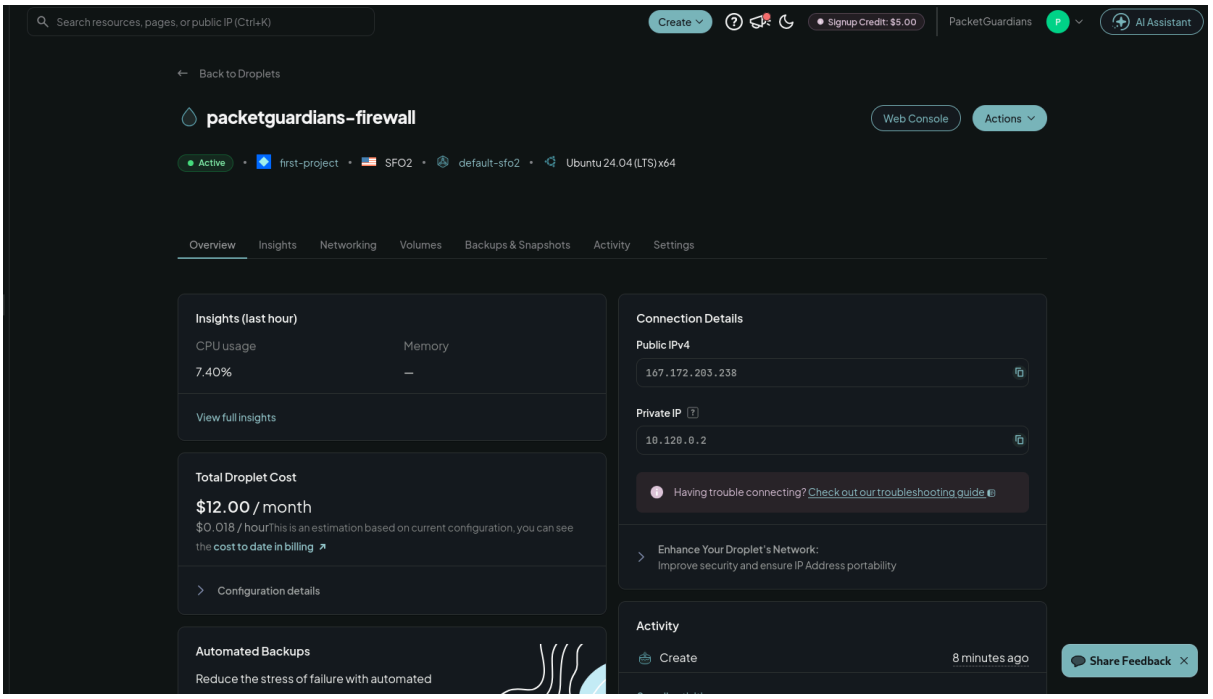
Here, I give the droplet a name that suits our project. At first, when making the project proposal in lab 7, I thought we needed a team name and came up with this funky one, so I decided to keep it here.

After putting payment method and creating droplet



Here, I am able to see my VPS being created. After creation, I should be assigned a public IPv4 and private IP address.

Droplet (VPS) information

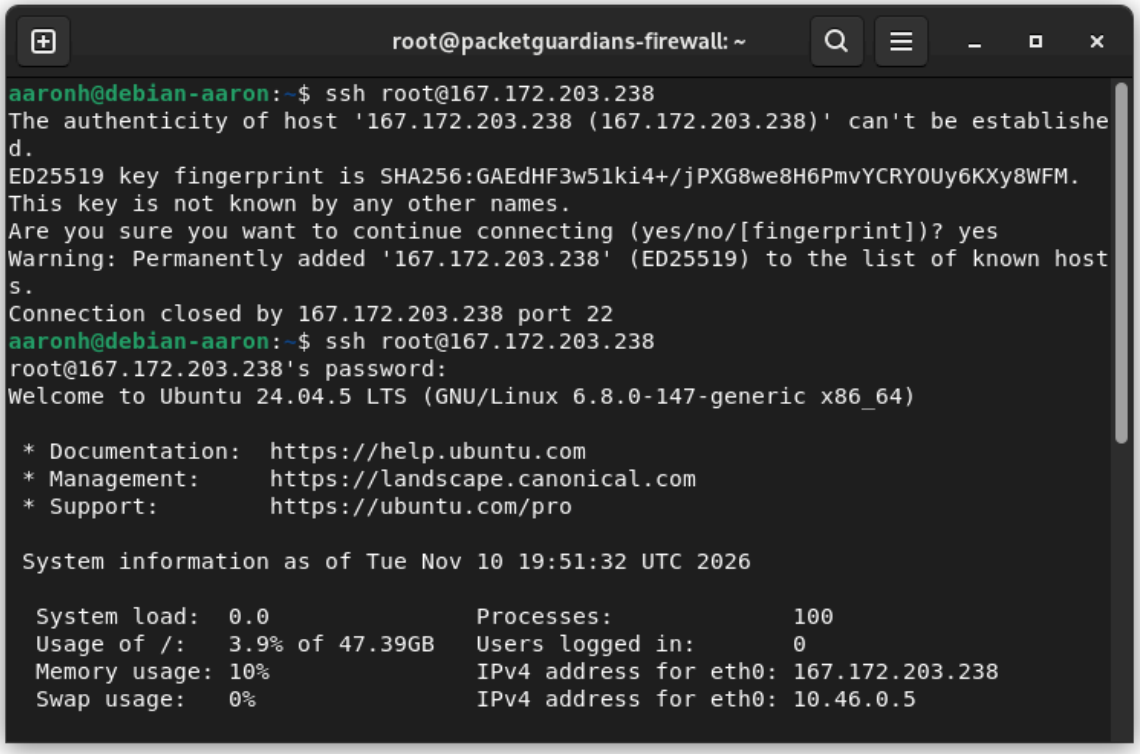


Here, the VPS is finally created. I was given an IPv4 address of: 167.172.203.238 and a private IP Address of: 10.120.0.2. The following is the full configuration of the droplet I created.

Provider: DigitalOcean
Plan: Basic, Regular CPU
Specifications: 1 vCPU, 2 GB RAM, 50 GB SSD
Price: 12$ USD
Datacenter: San Francisco (SFO2)
OS: Ubuntu 24.04 LTS x64
Public IPv4: 167.172.203.238
Private: 10.120.0.2
Authentication: SSH using a password
Status: Open and running

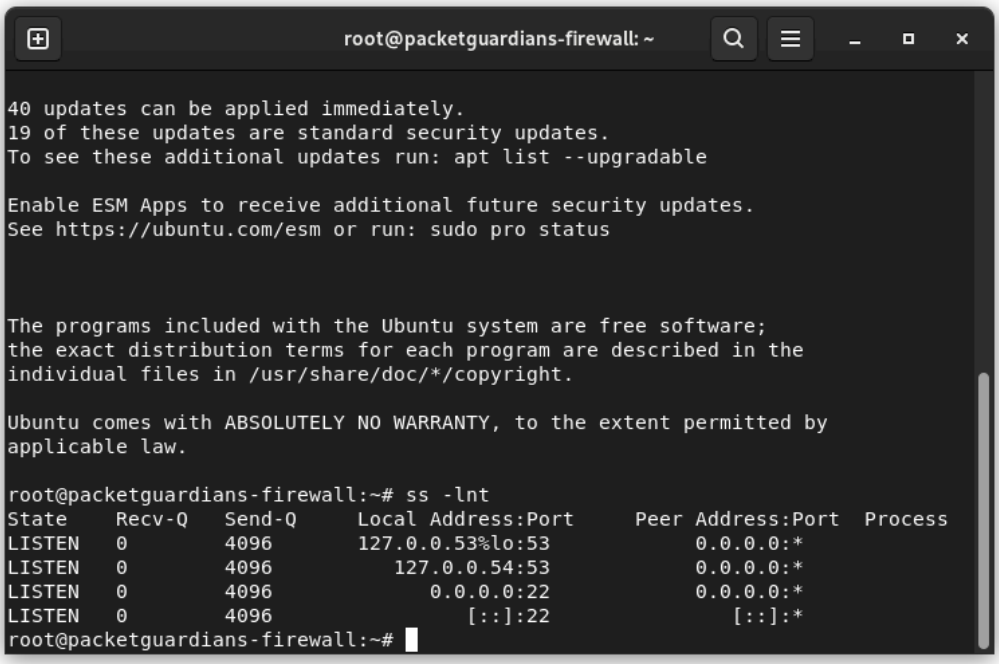
Testing SSH Access

To test remote access to the newly created droplet, I will connect from my Linux terminal using the password I created.



Here, I launched my first connection which needed a SHA256 fingerprint verification. After typing yes, I was prompted for my password. The first attempt took too long and failed because I was confused by the key fingerprint message. Running ssh root@167.172.203.238 again, I was prompted for a password (the one I created during the droplet process) and I was able to go into my VPS!

Running ss -lnt



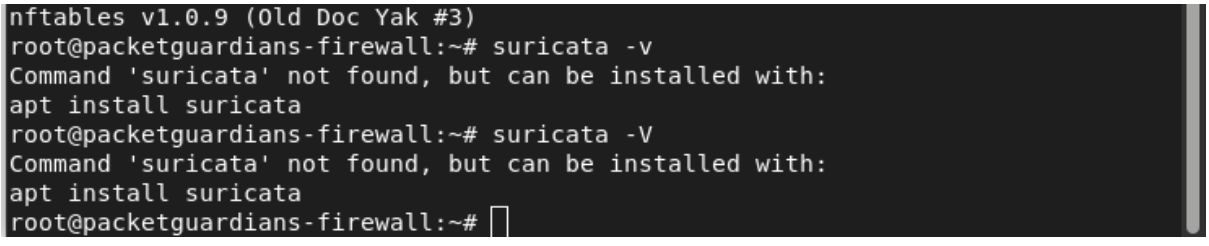
After running ss -lnt to check the servers listening ports, I see port 22 (for ssh) and port 53 (for dns) open.

Checking for all the software needed for our firewall

Usually, the linux distro should come with essential software already installed. To verify, I will check if it comes with nftables first.



I confirmed that nftables version 1.0.9 is already installed on the VPS.



Suricata is not installed, so I will do so by doing apt update first then apt install suricata.

```
Setting up libwww-perl (6.76-1ubuntu0.1) ...
Setting up oinkmaster (2.0-4.2) ...
Setting up liblwp-protocol-https-perl (6.13-1) ...
Processing triggers for libc-bin (2.39-0ubuntu8.9) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
 systemctl restart unattended-upgrades.service

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@packetguardians-firewall:~# suricata -V
This is Suricata version 7.0.3 RELEASE
root@packetguardians-firewall:~#
```

After running the command, I can verify Suricata version 7.0.3 is successfully installed.

```
root@packetguardians-firewall:~# systemctl is-active suricata
active
root@packetguardians-firewall:~# nft list ruleset
root@packetguardians-firewall:~#
```

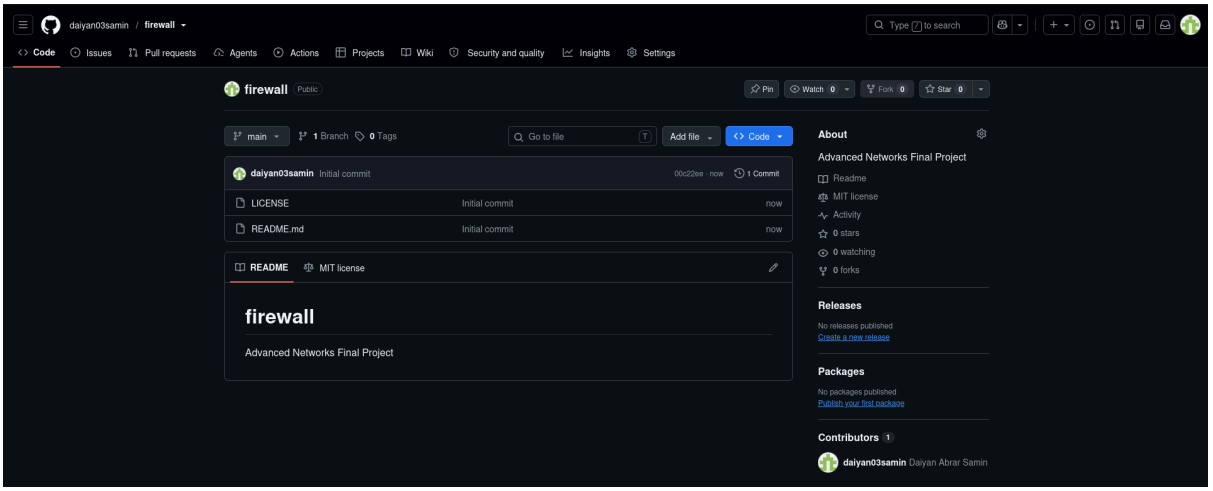
Here, I check to make sure the suricata service is active and ready for use.

And then I checked the nftables list of rules which was expectedly empty because we didn't make any rules yet, so everything is good.

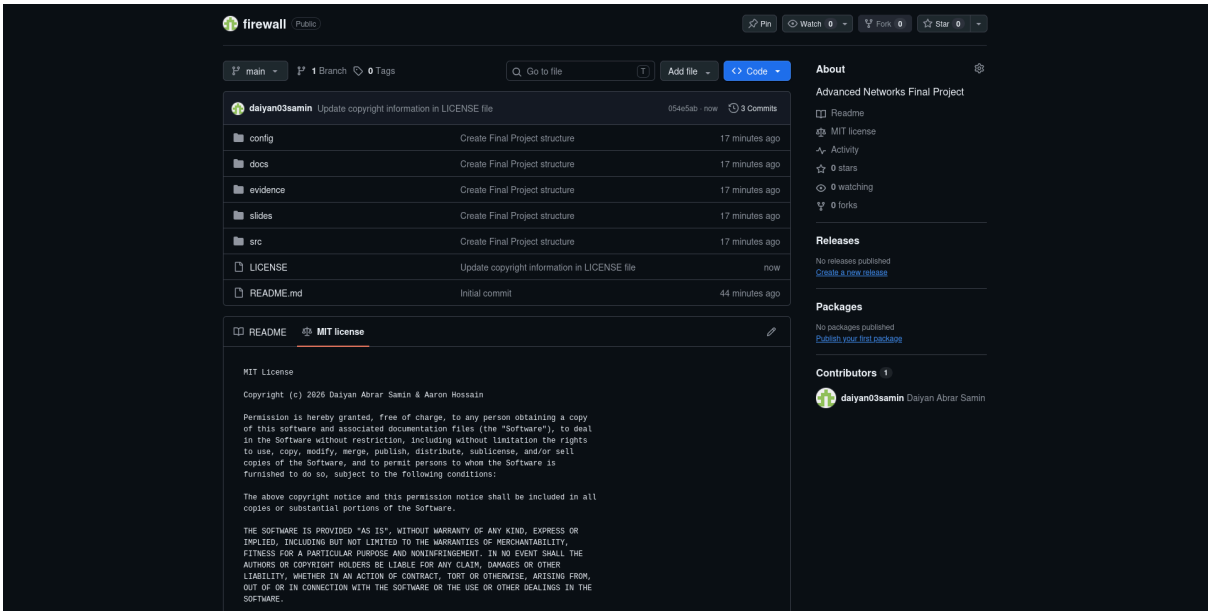
Now we have the VPS and all the software needed set up.

Daiyan’s Work

Repository Setup



Initial Github repository creation with README.md and MIT LICENSE and added Aaron as a collaborator.



I added a MIT license for our project.

```
daiyan@Debian:~$ sudo apt install git -y
[sudo] password for daiyan:
git is already the newest version (1:2.47.3-0+deb13u1).
The following package was automatically installed and is no longer required:
  linux-image-6.12.94+deb13-amd64
Use 'sudo apt autoremove' to remove it.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 44
daiyan@Debian:~$ git --version
git version 2.47.3
daiyan@Debian:~$ git clone https://github.com/daiyan03samin/firewall.git
Cloning into 'firewall'...
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (4/4), done.
daiyan@Debian:~$ cd firewall
daiyan@Debian:~/firewall$ ls -la
total 20
drwxrwxr-x  3 daiyan daiyan 4096 Nov 10 14:16 .
drwx------ 20 daiyan daiyan 4096 Nov 10 14:16 ..
drwxrwxr-x  8 daiyan daiyan 4096 Nov 10 14:16 .git
-rw-rw-r--  1 daiyan daiyan 1075 Nov 10 14:16 LICENSE
-rw-rw-r--  1 daiyan daiyan   43 Nov 10 14:16 README.md
daiyan@Debian:~/firewall$
```

Connect to github remote repo from command line

```
daiyan@Debian:~/firewall$ mkdir -p slides src config docs/report docs/plan evidence
daiyan@Debian:~/firewall$ touch slides/.gitkeep src/.gitkeep config/.gitkeep docs/report/.gitkeep
docs/plan/.gitkeep evidence/.gitkeep
daiyan@Debian:~/firewall$ git add .
daiyan@Debian:~/firewall$ git commit -m "Create Final Project structure"
[main d626c20] Create Final Project structure
 Committer: Daiyan Abrar <daiyan@debian>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

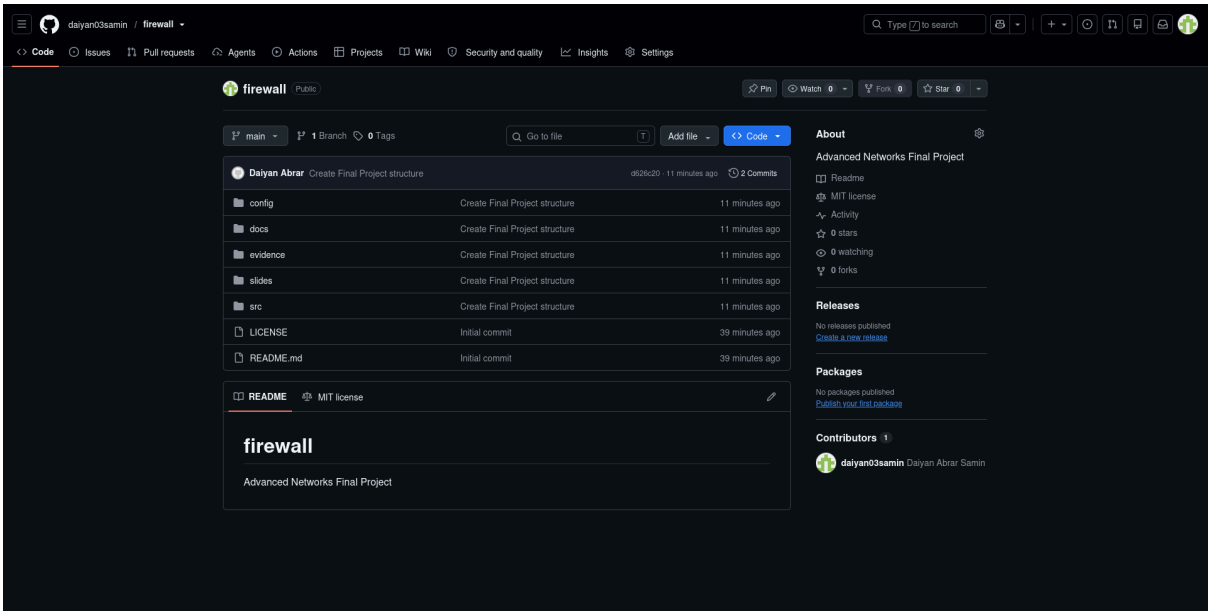
 6 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 config/.gitkeep
 create mode 100644 docs/plan/.gitkeep
 create mode 100644 docs/report/.gitkeep
 create mode 100644 evidence/.gitkeep
 create mode 100644 slides/.gitkeep
 create mode 100644 src/.gitkeep
```

Create the folders with .gitkeep empty files

```
daiyan@Debian:~/firewall$ git push
Username for 'https://github.com': daiyan03samin
Password for 'https://daiyan03samin@github.com':
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 20 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 470 bytes | 470.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/daiyan03samin/firewall.git
   00c22ee..d626c20  main -> main
daiyan@Debian:~/firewall$
```

Push the folders to remote repository

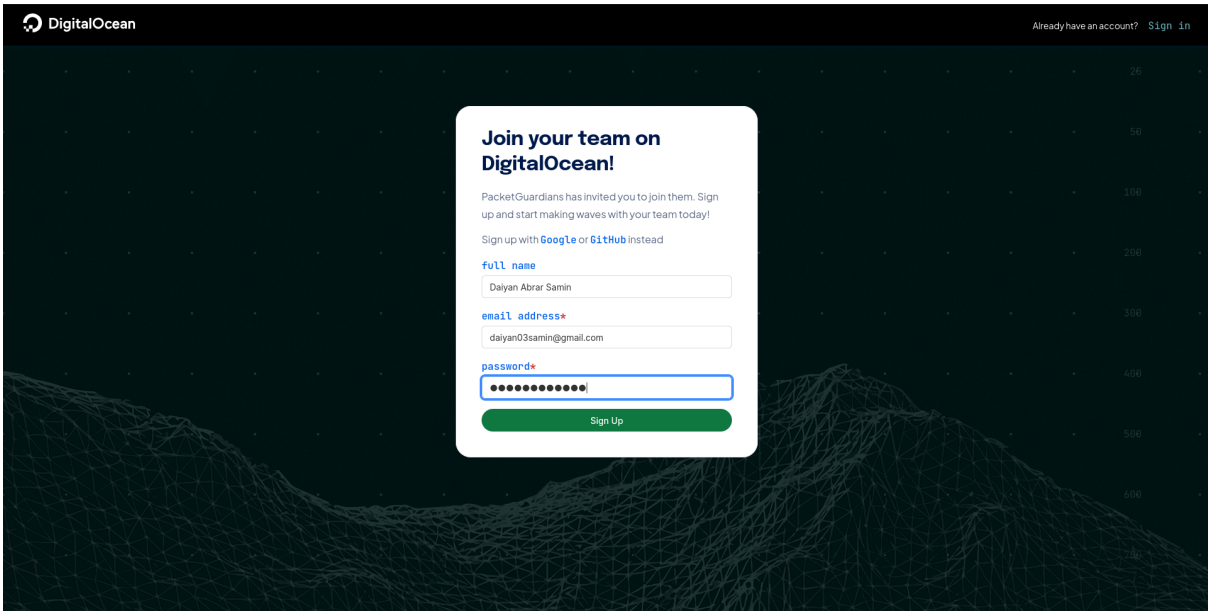
Organize the repo



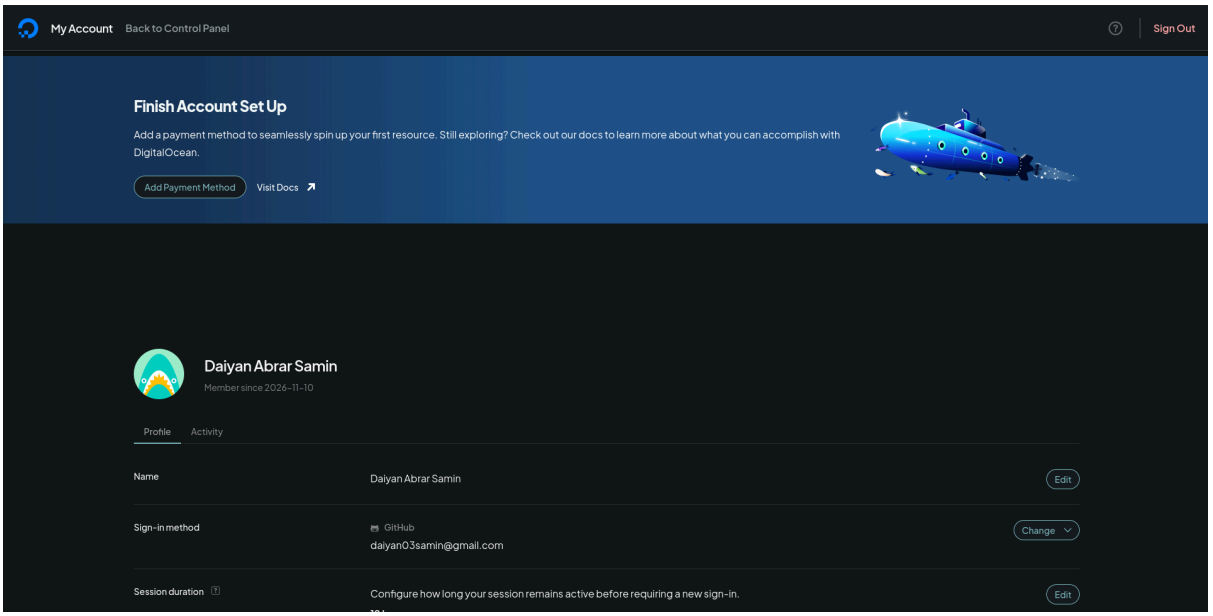
I set up the project folder structure in our github repo.

Aaron Hossain Daiyan Abrar Samin

Digital Ocean Account Setup

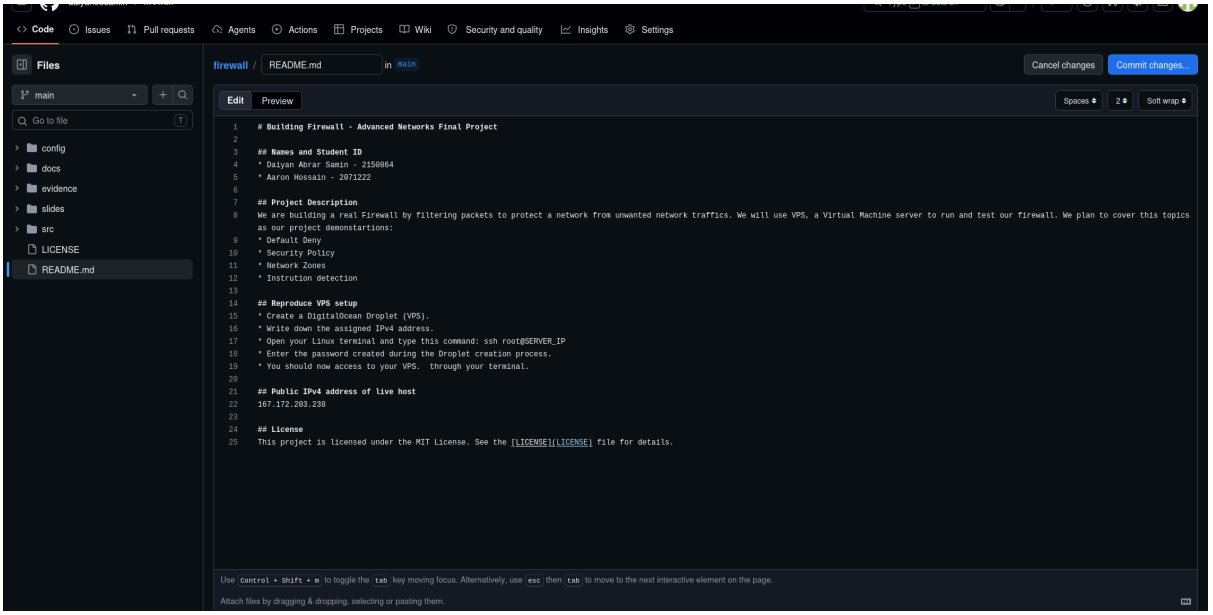


I created a Digital Ocean account using my GitHub account



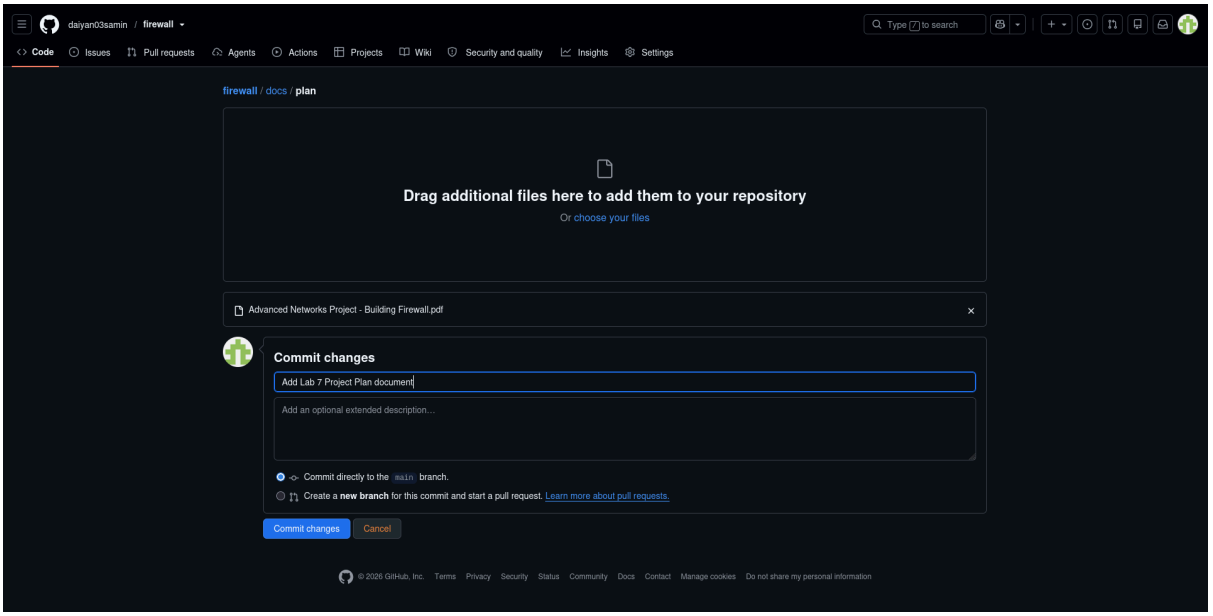
Aaron Hossain Daiyan Abrar Samin

Update the README file



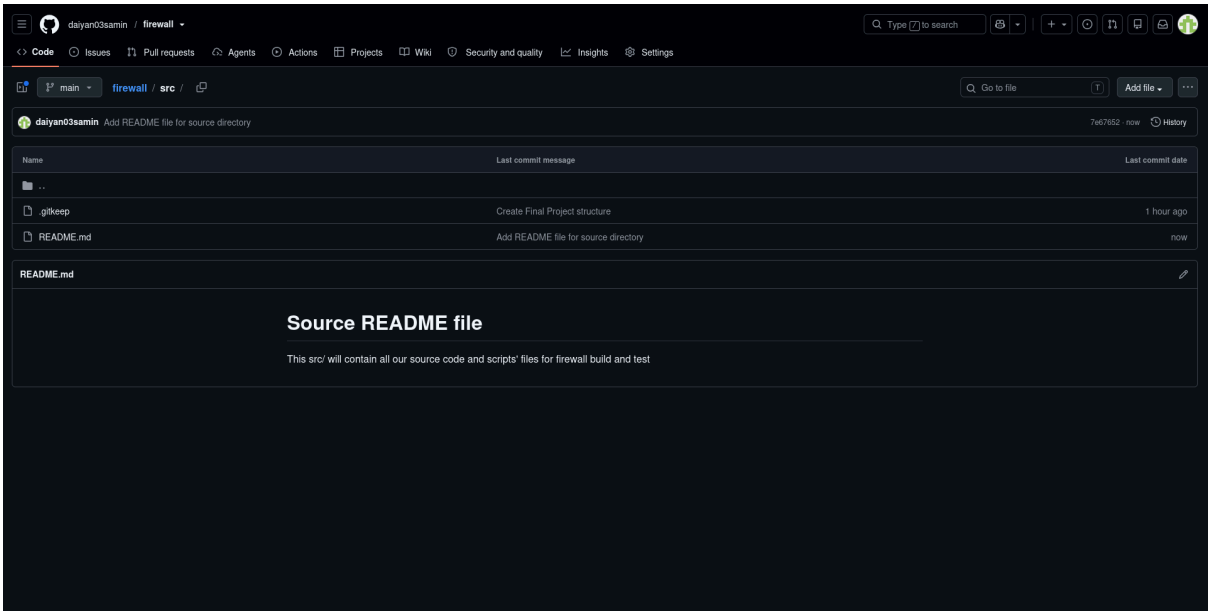
I updated the README file based on Lab 9 instructions.

Lab 7 plan add



Lab 7 Project plan pdf document has added.

src/ directory README file



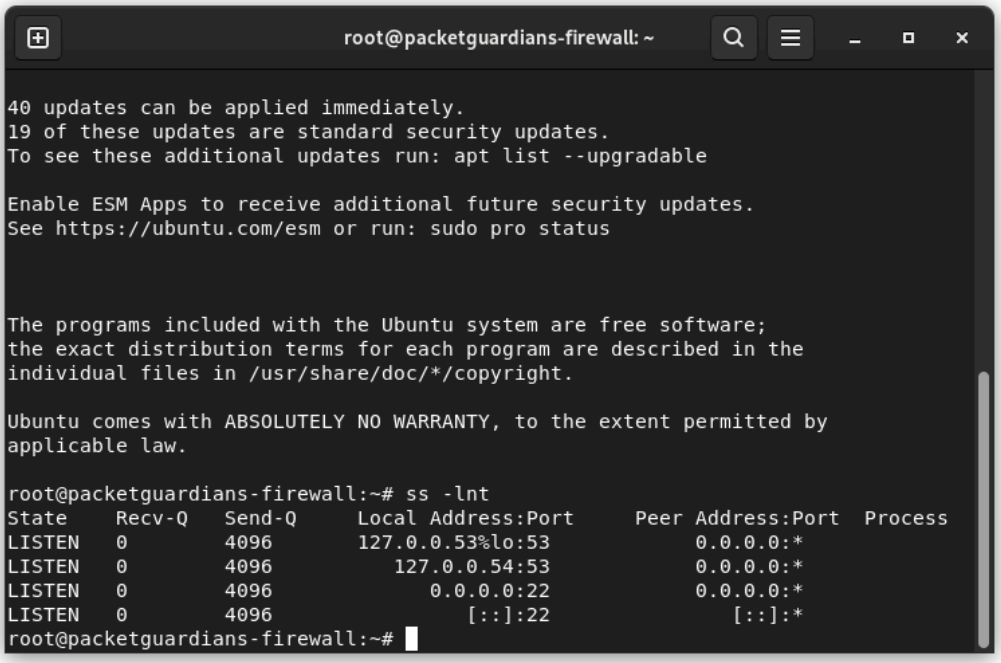
A short description file has added to src directory.

Hardware

Portable SSD - This hosts the Linux OS that I will use to send requests.
It is a Netac MobileDataStar 465.8G Usb. It is installed and on hand and ready for use.

Hosting

Provider: DigitalOcean
Plan: Basic, Regular CPU
Specifications: 1 vCPU, 2 GB RAM, 50 GB SSD
Price: 12$ USD
Datacenter: San Francisco (SFO2)
OS: Ubuntu 24.04 LTS x64
Public IPv4: 167.172.203.238
Private: 10.120.0.2
Authentication: SSH using a password
Status: Open and running



Software

Ubuntu Server

Version: 24.04.5 LTS
Command used: cat /etc/os-release
Role: OS hosting the firewall and IDS

Nftables (already installed)

Version: 1.0.9
Command used: nft –version
Role: Configure and enforce firewall rules

Suricata (had to install)

Version: 7.0.3
Command used: suricata -V
Role: Inspect traffic and generate alerts

We dropped the local VirtualBox setup with a DigitalOcean VPS, so VM is no longer needed.

Aaron Hossain Daiyan Abrar Samin

What ran today

Everything from setting up the repository to setting up the VPS was successful. The softwares needed (nftables and Suricata) were verified successfully. All the successes are shown in our work above.

Who did what today

Aaron: I created and set up the VPS and tested all the essential softwares are installed, set up and running. I also documented my entire process and did research with specific Linux commands such as actually connecting to the VPS through my Linux terminal using ssh root@serverip.

Daiyan: I created up our GitHub repository and set up the project directories, added and updated a README files. Finally I created DIGITAL Ocean account and get access to our project.

Who does what from now until the presentation

Since we are running out of time for the lab, we will discuss and confirm our tasks later.